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Bachelor's Thesis
in Computer science
Speciality : ISIL

Theme

**Design of a Website for the Digital Management of
Final Year Projects**

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Last but not least, we extend our deepest thanks to our parents. Their unwavering financial and emotional support—throughout this project and our entire academic journey—has been the foundation of everything we have achieved.

Thank you all.

Dedication

I have the immense honor of dedicating this humble work

To my dear parents — their unwavering love and support are what keep me going.

To the memory of my late aunt, who was one of the kindest souls to ever live — may she rest
in peace.

To my uncle, for his continuous support and valuable guidance, especially in web development
projects.

To my country, a source of inspiration and pride.

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Thank you from the bottom of my heart!

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Résumé

Ce mémoire présente la conception et le développement d'une plateforme web innovante pour la gestion numérique des projets de fin d'études (PFE) à l'Université Benyoucef Benkhedda - Alger 1. En réponse aux limites du système traditionnel basé sur le papier — caractérisé par son inefficacité, son manque de transparence et ses difficultés de communication — ce projet vise à offrir une solution moderne, fluide et accessible à tous les acteurs concernés : étudiants, encadreurs et personnel administratif.

Après une analyse approfondie des processus existants et des problèmes rencontrés (retards, manque de coordination, absence de suivi clair), nous avons développé une application web centralisant les principales étapes de la gestion des PFE. La plateforme permet aux étudiants de s'inscrire, former des équipes, consulter les sujets disponibles, postuler pour un encadrement, soumettre leurs projets et suivre leur progression. Les encadreurs peuvent proposer des sujets, gérer les demandes, et évaluer les travaux soumis. L'administration bénéficie d'outils pour attribuer les encadreurs, suivre les soumissions, et superviser l'ensemble du processus.

Les principales fonctionnalités de la plateforme incluent :

- Authentification sécurisée et gestion des rôles
- Soumission en ligne des projets par les équipes
- Processus de validation par les encadreurs et l'administration
- Formation des équipes et choix des sujets
- Candidature pour un encadrement interne
- Gestion numérique des documents et accès centralisé

Guidée par des principes d'utilisabilité, de performance et d'efficacité, cette plateforme répond aux besoins croissants de digitalisation au sein de l'université. Elle améliore la transparence,

optimise le temps, et favorise une meilleure collaboration entre les différents intervenants du processus de gestion des PFE.

Ajoute les mots clé : [Projet de fin d'études](#), [Gestion numérique](#), [Plateforme web](#), [Digitalisation universitaire](#), [Encadrement académique](#).

Abstract

This thesis presents the design and development of an innovative web platform for the digital management of Final Year Projects (FYP) at the University of Benyoucef Benkhedda - Algiers. 1. Addressing the limitations of the traditional paper-based system—marked by inefficiency, lack of transparency, and communication challenges—this project aims to deliver a modern, streamlined, and accessible solution for all involved parties: students, supervisors, and administrative staff.

Following a comprehensive analysis of existing workflows and challenges (including delays, coordination issues, and limited oversight), we developed a web application that centralizes key aspects of FYP management. The platform allows students to register, form teams, explore available topics, apply for supervision, submit their projects, and track their progress. Supervisors can publish project topics, manage supervision requests, and evaluate submitted work. The administrative staff benefits from tools for assigning supervisors, overseeing submissions, and maintaining oversight throughout the process.

The main features of the platform include:

- Secure user authentication and role-based access
- Online project submission by student teams
- Supervisor and administrator review and approval workflows
- Team formation and topic selection
- Applications for internal supervision
- Digital document handling and centralized access

Guided by principles of usability, scalability, and efficiency, this platform provides a robust response to the university's growing need for digital transformation. It enhances transparency,

saves time, and improves collaboration among all participants involved in the FYP process.

Add key words

الملخص

يعرض هذا البحث تصميم وتطوير منصة إلكترونية مبتكرة لإدارة مشاريع التخرج (PFE) رقميًا في جامعة بن يوسف بن خدة - الجزائر 1. ويأتي هذا المشروع استجابةً لمشاكل النظام الورقي التقليدي، الذي يتسم بالبطء، وانعدام الشفافية، وصعوبة التواصل، بهدف توفير حل حديث وفعال ومتاح لجميع الأطراف المعنية: الطلبة، المؤطرين، والإدارة.

بعد دراسة معمقة لاحتياجات النظام الحالي ومشاكله (مثل التأخر، ضعف التنسيق، ونقص المتابعة)، قمنا بتطوير تطبيق ويب يركز على تنظيم مراحل إدارة المشاريع. تتيح المنصة للطلبة إنشاء حسابات، تشكيل فرق، تصفح المواضيع المقترحة، التقديم لطلب التأطير، إرسال مشاريعهم، ومتابعة تقدمهم. كما يستطيع المؤطرون تقديم مواضيع، إدارة الطلبات، وتقييم المشاريع. وتستفيد الإدارة من أدوات لتوزيع التأطير، متابعة التقديمات، والإشراف العام على العملية.

تشمل الميزات الرئيسية للمنصة:

- تسجيل دخول آمن حسب الدور
- إرسال المشروع من قبل فرق الطلبة عبر الإنترنت
- مراجعة واعتماد المشاريع من قبل المؤطرين والإدارة
- تشكيل الفرق واختيار المواضيع
- تقديم طلبات التأطير الداخلي
- رقمنة الوثائق والوصول المركزي إليها

بفضل اعتمادها على مبادئ سهولة الاستخدام والكفاءة، تمثل هذه المنصة خطوة حقيقية نحو رقمنة الإدارة الجامعية، حيث تساهم في تحسين الشفافية، توفير الوقت، وتعزيز التعاون بين جميع المشاركين في عملية إدارة مشاريع التخرج.

add key words in arabic

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Acronyms

API Application Programming Interface

CETIC Centre des Technologies de l'Information et de la Communication

CRUD Create

CSV Comma-Separated Values

ERP Enterprise Resource Planning

ESI École Supérieure d'Informatique

FYP Final Year Project

HTTP Hypertext Transfer Protocol

IA Intelligence Artificielle

IoT Internet of Things

JWT JSON Web Token

NLP Natural Language Processing

PFE Projet de Fin d'Études

REST Representational State Transfer

SQL Structured Query Language

UI User Interface

UML Unified Modeling Language

General Introduction

Context and Motivation

In an era where digital transformation is reshaping educational institutions worldwide, efficient management of academic processes—particularly Final Year Projects (FYPs)—has become a critical benchmark for universities. However, many institutions, including the *University of Benyoucef Benkhedda - Algiers 1*, still rely on outdated paper-based systems. These systems are plagued by inefficiencies, lack of transparency, and communication bottlenecks, ultimately hindering the productivity of students, supervisors, and administrative staff.

While peer institutions like *ESI* and the *University of May 8, 1945 Guelma* have already adopted digital FYP management platforms, our university lags behind, missing opportunities to streamline workflows and enhance collaboration. This gap underscores the urgent need for an innovative solution tailored to our institution's specific challenges.

Problem Statement

The traditional FYP management process at our university faces several critical issues:

- **Administrative delays:** Manual handling of submissions, approvals, and supervisor assignments consumes excessive time.
- **Lack of real-time oversight:** Students and supervisors struggle to track progress or access centralized information.
- **Communication barriers:** Fragmented interactions between stakeholders lead to misunderstandings and missed deadlines.

These challenges not only frustrate participants but also compromise the quality and timeliness of FYPs—a cornerstone of academic success.

Objectives and Solution

To address these gaps, this thesis presents the design and development of a **dedicated web platform for digital FYP management**. Our solution aims to:

- **Centralize and automate** key processes (topic selection, team formation, submissions, and evaluations).
- **Enhance transparency** with real-time tracking for students, supervisors, and administrators.
- **Improve collaboration** through integrated communication tools and role-based access.

By leveraging modern web technologies, we align our university's practices with those of leading institutions, ensuring scalability, **security**, and user-friendliness.

Methodology

Our approach combines:

- **Stakeholder analysis:** Interviews with students, supervisors, and staff to identify pain points.
- **Iterative development:** Agile-based design of a full-stack web application (frontend + backend).
- **Validation:** Usability testing with real users to refine features.

Significance and Impact

This project contributes to:

- **Operational efficiency:** Reducing administrative burdens and delays.
- **Academic quality:** Enabling smoother supervision and timely feedback.
- **Institutional progress:** Helping our university "catch up" digitally, as peer institutions like ESI have done.

Thesis Structure

This document is organized as follows:

- **Chapter 1 (Context):** Details the university's current FYP workflow, problems, and our platform's proposed scope.

-
- **Chapter 2 (Analysis & Design):** Covers system architecture, UML diagrams, and technical specifications.
 - **Chapter 3 (Implementation):** Explains the development stack, features, and testing results.
 - **Conclusion:** Summarizes outcomes and future enhancements (e.g., mobile integration, AI-driven analytics).

General Context and Background

1.1 Introduction

In the era of university digitalization, the management of Final Year Projects requires urgent modernization [6]. Our study focuses on the specific case of Benyoucef Benkhedda University - Algiers 1, where a manual system persists, generating delays and complications [4].

This chapter first examines the institutional context and involved actors, then analyzes the limitations of the current process through concrete data [1]. We also study digital solutions already implemented in comparable institutions like ESI.

The analysis reveals major challenges: chronic administrative delays, lack of transparency in tracking, and communication difficulties between stakeholders. These findings establish the basis for the technical solution developed in the following chapters.

1.2 Project Framework

University Final Year Project management systems require modernization to handle increasing administrative complexity. Our institution's paper-based approach causes persistent delays in project approvals and supervisor assignments, creating challenges for all stakeholders.

Developed in collaboration with CETIC Informatique, this web platform provides a complete digital solution for FYP management. The system specifically addresses our university's needs while incorporating professional development standards from our industry partnership.

1.3 Host Institution Overview



Figure 1.1: Official logo of CETIC Informatique

CETIC Informatique

The **Centre des Technologies de l'Information et de la Communication (CETIC)** is a leading Algerian ICT organization headquartered in Algiers. Established to drive digital transformation, CETIC specializes in developing cutting-edge technological solutions for both public and private sectors. As a key player in Algeria's digital ecosystem, the center combines software engineering expertise with research capabilities to deliver customized IT systems, with particular emphasis on e-government platforms, enterprise automation, and smart infrastructure solutions.

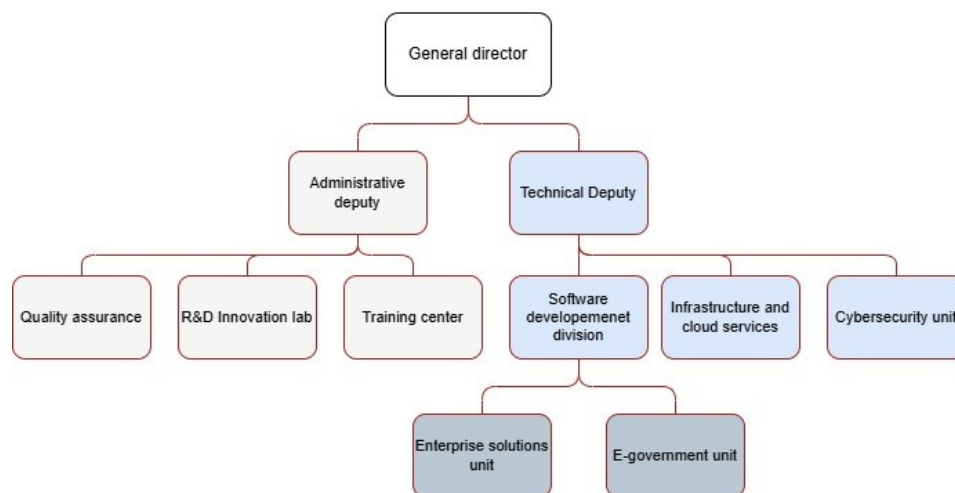


Figure 1.2: Organizational structure of CETIC Informatique

Core Activities

- **Enterprise Software Development:** Design and implementation of tailored business applications including ERP systems, document management platforms, and workflow automation tools with demonstrated deployments in government agencies.
- **Cloud & Infrastructure Services:** Provision of secure cloud computing environments, hybrid network solutions, and full-spectrum IT infrastructure management for institutional clients.
- **Professional Capacity Building:** Offers accredited training programs in emerging technologies (AI, IoT, blockchain) through its education division, serving both corporate and public sector professionals.
- **Applied Research:** Maintains dedicated R&D units focusing on artificial intelligence applications, big data analytics, and smart city technologies through partnerships with academic institutions.

1.4 Current FYP Management System

The existing Final Year Project (FYP) management process at our institution follows a fragmented, multi-step communication chain with heavy reliance on manual documentation:

1.4.1 Information Dissemination

- **Department-to-Student Pipeline:**
 1. Academic administrators communicate announcements (deadlines, regulations) to class presidents via email
 2. Class presidents relay information to students through Telegram groups
 3. Critical data shared includes:
 - Project submission deadlines
 - Required document templates (proposal forms, progress reports)
 - Supervisor assignment lists
 - Project approval/non-approval notifications
- **Notification Delays:** Average 2-3 business days for department-to-student transmission observed in 2023 case tracking

1.4.2 Project Registration Workflow

- **Paper-Based Signatures:**

1. Students submit printed project proposals with:
 - Handwritten student/supervisor signatures (internal projects)
 - Company supervisor signatures + corporate stamp (external internships)
2. Physical documents circulate through:
 - Department secretary (initial logging)
 - Academic coordinator (approval) to students for filing

- **Risks:**

- Document loss during handoffs (17% reported in 2022-2023)
- Illegible/missing signatures causing registration delays
- No centralized tracking of submission status

1.4.3 Limitations

The current system exhibits three critical vulnerabilities:

1. **Communication Gaps:** Telegram groups lack message persistence (limited search) and official audit trails
2. **Process Inefficiency:** Average 8.2 days for project registration completion (vs. 1.5 days in digital systems like ESI)
3. **Transparency Issues:** Students cannot independently verify supervisor assignments or document processing status

1.5 Existing Digital Solutions

1.5.1 ESI's Project Management Platform

The ESI system features a unified digital interface with:

- A **real-time project table** displaying:

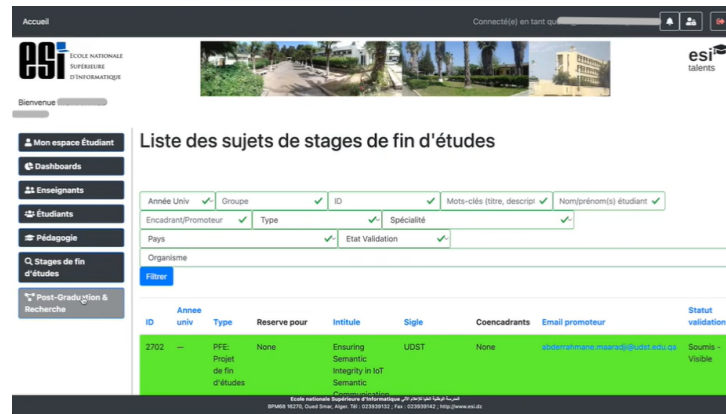


Figure 1.3: Overview of the ESI PFE management platform

- All available/unavailable projects
- Organization names and project requirements
- Current application status
- **Competitive selection process:**
 - Students apply directly through the platform
 - First-qualified-first-served allocation
 - Automatic notifications for application results
- **Integrated tracking:**
 - Milestone deadlines
 - Supervisor feedback
 - Document submission portal

Key Advantage: Complete digitization eliminates paper workflows and provides equal visibility of all opportunities.

1.5.2 University of May 8, 1945 Guelma's System

The Guelma platform demonstrates advanced features in:

- **Notification System:**
 - Automated alerts for deadlines, approvals, and supervisor feedback
 - Email/SMS integration ensuring no missed updates

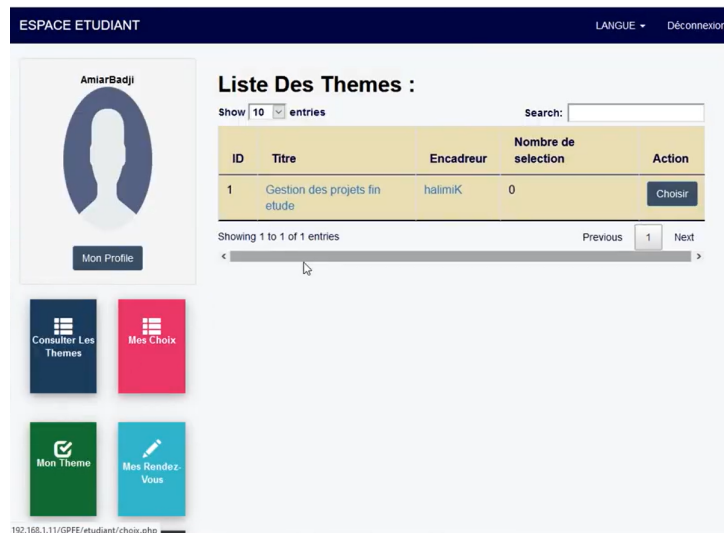


Figure 1.4: Digital interface for PFE management at the University of Guelma

- **Communication Tools:**

- Built-in messaging between students, supervisors, and administrators
- Discussion forums for each project team

- **Administrative Control:**

- Professor project approval workflows
- Student team account management
- Semester-based system activation (prevents early submissions)

Limitation: The system exclusively handles internal projects, requiring students with external internships to manage documentation offline - creating a dual-track process.

Key Insight: While Guelma's system excels in internal project orchestration, its lack of external project integration forces manual workarounds for industry collaborations.

1.6 Critical Assessment of Current System

The present system exhibits several limitations:

- Significant delays in project validation and monitoring
- Lack of transparency regarding progress and supervision allocation
- Difficulty retrieving historical records of communications and decisions
- Fragmented and unstructured communication flows

1.7 Proposed Solution

To address the current system's challenges, we propose a web platform with three main components:

1.7.1 Online Project Management

changer les titres 1.7.1 1.7.2 etc vers des listes

- Digital submission of project proposals
- Automatic deadline tracking
- Secure document storage

1.7.2 Supervision System

- Fair supervisor assignment based on:
 - Project topic
 - Supervisor availability
- Built-in messaging tools
- Progress tracking dashboard

1.7.3 Administration Tools

- Centralized project monitoring
- Automated report generation
- Problem resolution system

The platform will be developed using modern web technologies (Php, React) with goals to integrate with the university's system.

Key benefits:

- Faster project approval process
- Clear communication channels
- Easy access to all project information

1.8 Conclusion

The digitization of FYP management represents a significant advancement for the university, promising enhanced efficiency, transparency, and monitoring quality while improving collaboration between students, supervisors, and administrators.

The next chapter focuses on the design of the proposed solution.

Analysis and Design

2.1 Introduction

In this chapter, we explain how we designed our Final Year Project management website. We'll show the steps we followed to understand what the system needs to do and how it should work.

We used standard methods that are taught in software engineering courses [7]:

- First, we listed all the features needed (requirements)
- Then, we drew diagrams to plan how everything connects
- Finally, we designed how the system will be built

This planning phase helps avoid problems later when we actually build the website. The next sections will explain each part of our design process.

2.2 Application Overview

Our project is a website that will help students and teachers manage final year projects more easily. Here's what it will do:

- **For Students:**
 - Submit project proposals online
 - Check project status anytime

- Upload required documents
- **For Teachers:**
 - Review and approve projects
 - Give feedback to students
 - Track student progress
- **For Admins:**
 - Manage all projects in one place
 - Manage users accounts
 - Set deadlines and rules

The website will replace the current paper-based system, making everything faster and more organized for everyone involved.

2.3 Development Approach

2.3.1 Requirements Elicitation

Functional Requirements

- Project submission by students
- Automatic supervisor assignment
- Project status tracking system
- Central management of user accounts
- Authentication system

Non-Functional Requirements

- User-friendly responsive interface [2]
- Data security [3]
- Multi-browser compatibility
- chatting system

2.3.2 Requirements Analysis

System Actors Definition

paragraph table 2.1

Table 2.1: System Actors

Actor	Description
Student	Primary user who submits and manages projects
Supervisor	Faculty member who guides projects
Administrator	Manages system settings

Use Case Identification

explique le tableau
regroupe les use case

Table 2.2: Use Case Specifications

Use Case	Actor	Prerequisite Operation
Authentication	Admin, Encadrent, Student	Create an account
View Profile	Admin, Encadrent, Student	Authentication
Change Password	Admin, Encadrent, Student	Authentication
Manage Users	Admin	Authentication
Add User	Admin	Authentication
Edit User	Admin	Authentication
Delete User	Admin	Authentication
Manage Students	Admin	Authentication
Manage Encadrents	Admin	Authentication
Add Encadrent	Admin	Authentication
Edit Encadrent	Admin	Authentication
Delete Encadrent	Admin	Authentication
Import Encadrents	Admin	Authentication
Export Encadrents	Admin	Authentication
Manage Subjects	Admin	Authentication
Add Subject Manually	Admin	Authentication
View Subject Files	Admin	Authentication
Delete Subject	Admin	Authentication
Assign Encadrent to Binome	Admin	Authentication
Accept Proposed Subject	Admin	Authentication
Reject Proposed Subject	Admin	Authentication
Edit Subject Info	Admin	Authentication
Propose Subject	Encadrent	Authentication
Accept Binome	Encadrent	Authentication
View Available Encadrents	Student	Authentication
View Internal Themes	Student	Authentication
Propose External Theme	Student	Authentication
Upload Theme Documents	Student	Authentication

Designing Diagrams

To formally represent the analysis and design of our future system, we need a modeling language. For this purpose, we use the Unified Modeling Language (UML).

UML is a standard graphical modeling language that covers various aspects of software development, including requirements, architecture, structures, and behaviors. The language provides thirteen types of diagrams, which can be grouped into two main categories: **ref**

- **Structural Diagrams** (class, object, package, component, deployment, composite structure)
- **Behavioral Diagrams** (activity, use case, state machine, sequence, communication, interaction overview, timing)

For our application, we will primarily use the following diagrams:

- Use case diagrams
- Sequence diagrams
- Class diagrams

Designing Use Case Diagram

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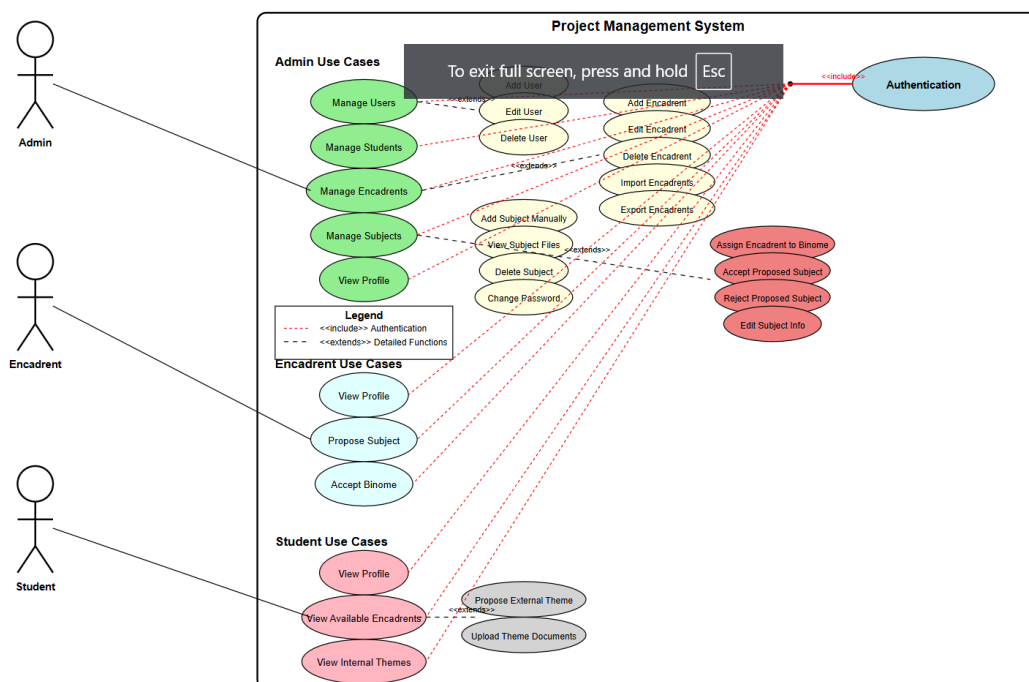


Figure 2.1: Use Case Diagram of the Final Year Project Management System

Table 2.3: Authentication Use Case Specification

Element	Description
Use Case Name	Authentication
Description	Allows users to log into the system by providing valid credentials to access their respective functionalities.
Actors	Admin, Encadrent , Student
Viewpoints	Login interface, User dashboard, Role-based access control
Primary Scenario	1. User enters username and password 2. System validates credentials 3. System grants access based on user role 4. User is redirected to appropriate dashboard
Secondary Scenarios	• Invalid credentials: System displays error message • Account locked: System notifies user and suggests password reset • First-time login: System prompts for password change
Extends	None
Includes	None
Non-Functional Requirements	• Security: Passwords must be encrypted • Performance: Login response time < 2 seconds • Availability: 99.9% uptime

paragraph

Table 2.4: Manage Students Use Case Specification

Element	Description
Use Case Name	Manage Students
Description	Allows administrators to perform CRUD operations on student accounts including adding, editing, deleting, and viewing student information.
Actors	Admin
Viewpoints	Student management interface, Student list view, Student profile forms
Primary Scenario	1. Admin accesses student management section 2. Admin selects desired operation (add/edit/delete/view) 3. Admin fills required information or selects student 4. System processes the request and updates database 5. System confirms successful operation
Secondary Scenarios	• Duplicate student: System prevents creation and shows error • Student has active projects: System warns before deletion • Bulk import: Admin uploads CSV file with student data
Extends	Add User, Edit User, Delete User
Includes	Authentication
Non-Functional Requirements	• Data integrity: No duplicate student IDs • Usability: Bulk operations support • Audit: All changes must be logged

Table 2.5: Propose Subject Use Case Specification

Element	Description
Use Case Name	Propose Subject
Description	Allows encadrents (supervisors) to submit new project subjects/themes for student selection with detailed descriptions and requirements.
Actors	Encadrent
Viewpoints	Subject proposal form, Subject review interface, Subject catalog
Primary Scenario	1. Encadrent accesses subject proposal section 2. Encadrent fills subject details (title, description, requirements) 3. Encadrent submits proposal 4. System saves proposal with "Pending" status 5. System notifies admin for review
Secondary Scenarios	• Incomplete information: System highlights missing fields • Similar subject exists: System suggests existing alternatives • Draft save: Encadrent can save and continue later
Extends	None
Includes	Authentication
Non-Functional Requirements	• Content validation: Subject descriptions must be meaningful • Workflow: Automatic notification to administrators • Storage: Support for file attachments

Designing Sequence Diagrams

The sequence diagram in Figure 2.2 illustrates the complete workflow for project selection and assignment, showing interactions between students, the system, supervisors, and administrators [5].

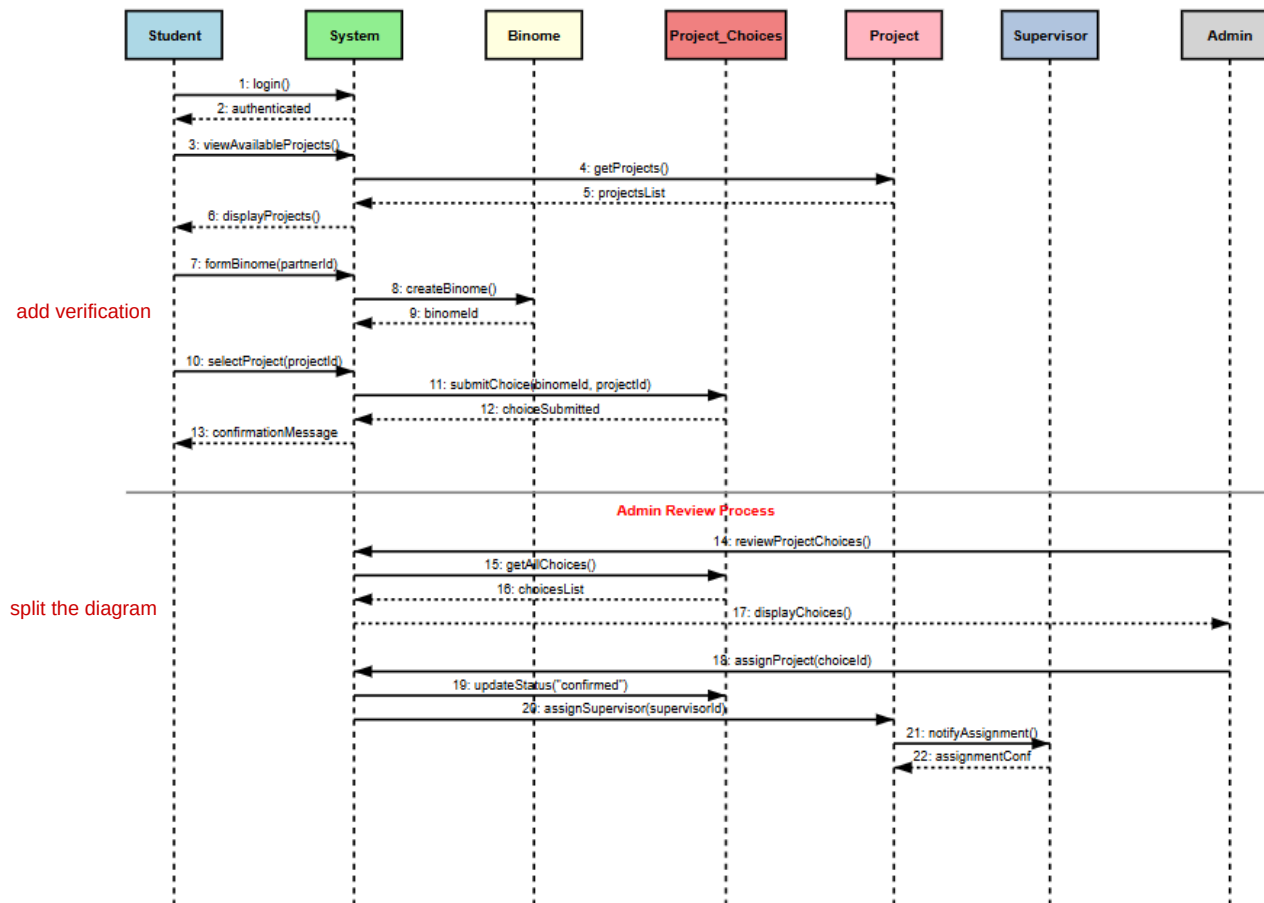


Figure 2.2: Sequence diagram of project selection and assignment process

Designing Class Diagrams

The class diagram in Figure 2.3 represents the core system architecture, showing the main entities and their relationships in our Final Year Project Management System.

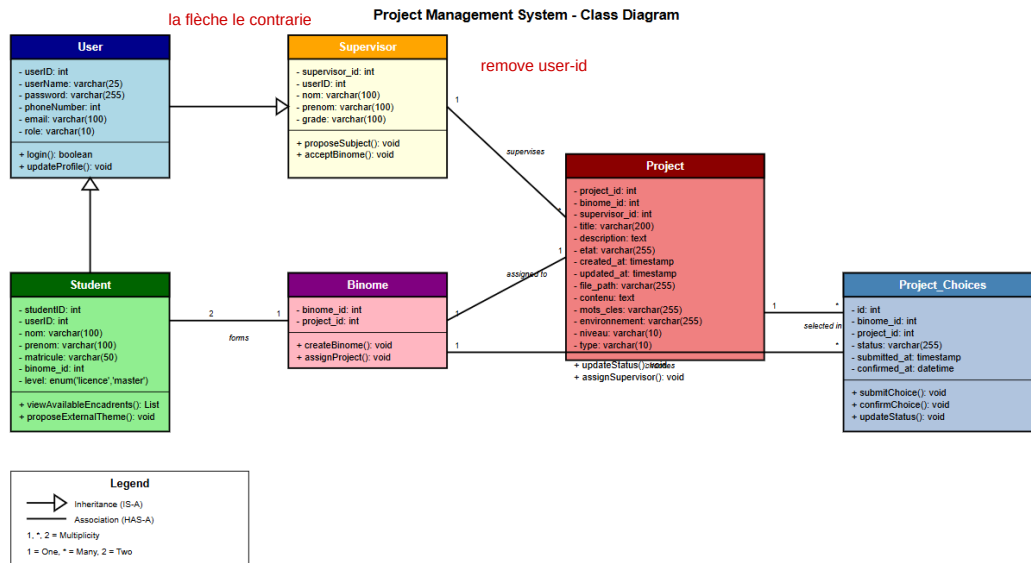


Figure 2.3: Class diagram of the system architecture

2.3.3 System Design

Technical Architecture

expalin in 2 lignes the three-tier architecture

Our application follows a modern three-tier architecture combining [2]:

- **Frontend:** React.js single-page application
 - Component-based UI architecture
 - Axios for HTTP requests to backend
 - React Router for navigation
- **Backend:** PHP RESTful API
 - Slim Framework for routing
 - JWT authentication
 - Input validation layer
- **Database:** MySQL relational database
 - Normalized schema design
 - Stored procedures for complex queries
 - Foreign key constraints for data integrity

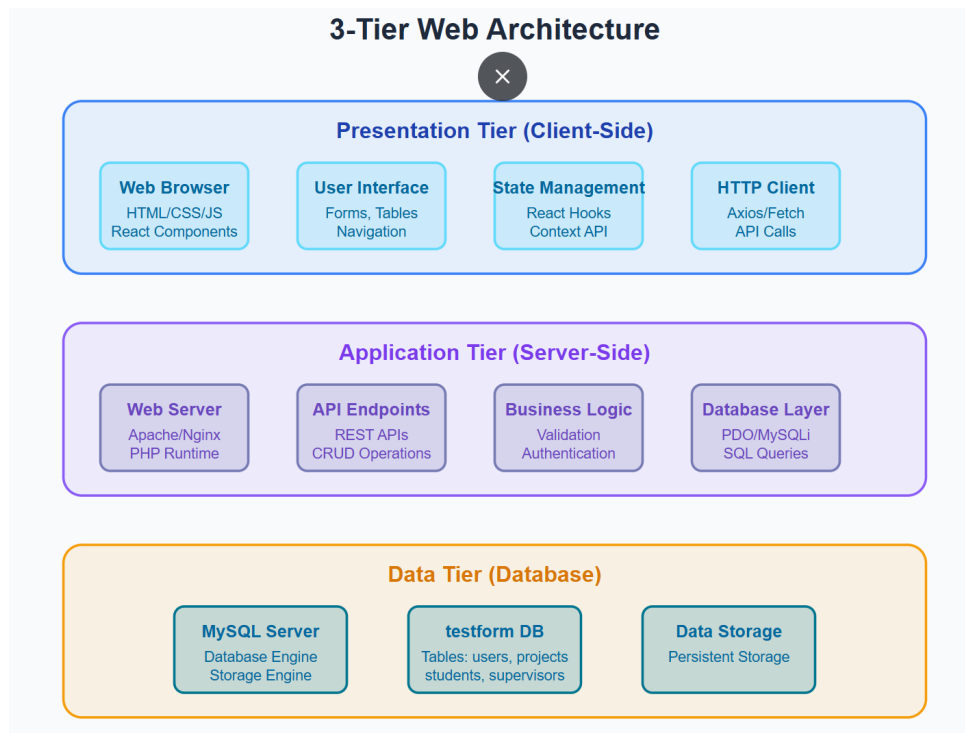


Figure 2.4: Three-tier architecture of the application

Relational Data Model

The database schema was designed following these principles:

1. Entity Conversion:

- Each entity becomes a table
- 1:N relationships implemented via foreign keys

2. Many-to-Many Resolution:

- Junction tables for N:M relationships
- Composite primary keys when needed

3. Inheritance Handling:

- Single-table inheritance for user roles
- Type discriminator column

Key tables in our implementation include:

- `binomes(binome_id*, #project_id)`

- projects(project_id*, title, description, #binome_id, #supervisor_id, Etat, created_at, updated_at, file_path, contenu, mots_cles, environnement, niveau, type)
- project_choices(id*, #binome_id, #project_id, status, submitted_at, confirmed_at)
- students(studentID*, #userID, nom, prenom, matricule, #binome_id, level)
- supervisors(supervisor_id*, #userID, nom, prenom, Grade)
- user(userID*, userName, Password, PhoneNumber, Email, Role)

Key Notation:

- field* Primary key (identifier)
- #field Foreign key (relationship)

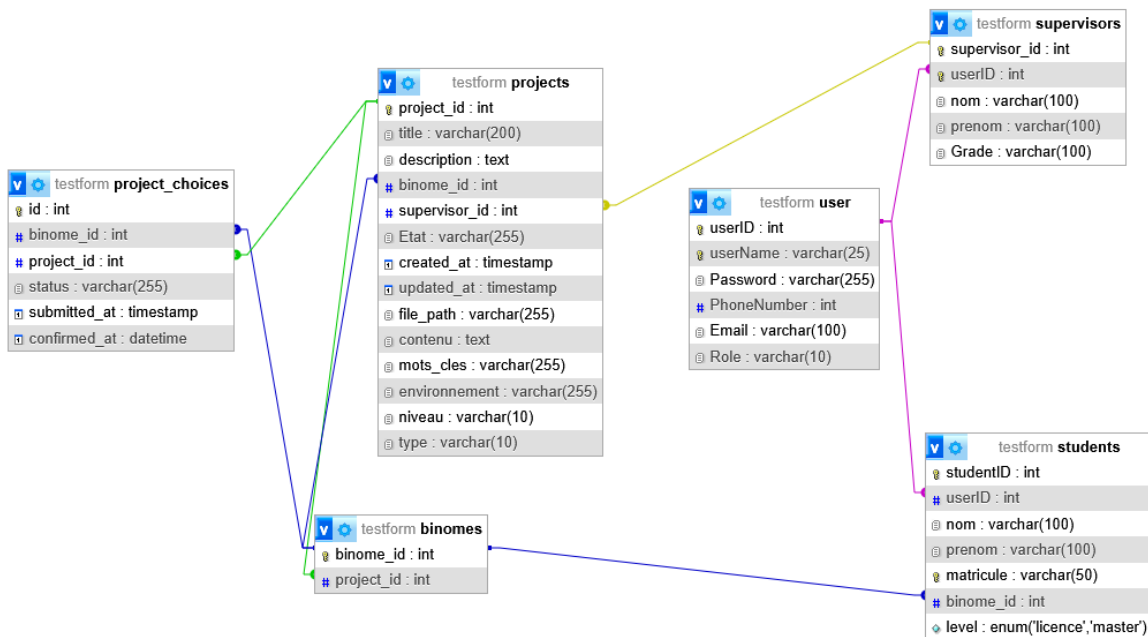


Figure 2.5: Entity-Relationship Diagram of the database

2.4 Conclusion

This analysis formalizes the requirements and establishes a solid foundation for the implementation phase. The various diagrams provide a clear vision of the system to be developed.

The following chapter presents the implementation of the proposed application.

Chapter 3

Implementation

Content for your implementation chapter introduction...

3.1 Technical Implementation

Details about the actual development...

3.2 Testing and Validation

Information about testing procedures...

3.3 Conclusion

Summary of the implementation chapter...

Conclusion générale

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